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The Discovery of Agile Service Networks through the Use of Social Network Analysis

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Abstract – *There is a need to address the significant gap in our ability to measure and monitor the Key Performance Indicators (KPIs) across service networks. The unprecedented growth in service-based business processes over a short period of time has underscored the need for understanding the mechanisms and theorising the business models and business process management adopted across many organisations today. This research presents a survey of the literature and argues that the inability of current Business Process Management (BPM) techniques to visualise and monitor web-enabled business processes prevents us from transforming information on network activity and infrastructures. This inhibits managers in anticipating change and adapting to more agile business practices in service science. Thus, this research-in-progress sets out to propose the need to develop a framework to enhance a manager's ability to monitor key performance indicators (KPIs) while improving business process restructuring practices through social network analysis (SNA).*

Keywords: *service science, business process management, agile service networks, key performance indicators, social network analysis.*

I. INTRODUCTION

This paper sets out to explore the literature on our ability to measure IT-enabled business processes and identify where weaknesses exist. Our traditional understandings of the 'organisation', with solid boundaries and internally focused operations, time, and individuality are becoming less apparent today. As competitive advantages of single organisational strategies continue to erode over recent years, organisations are experiencing greater demands to operate with increased innovation, collaboration, scalability, agility, and virtuality. Organisations must develop new initiatives to operate under these conditions and enhance their competitive edge. Information and communication technology (ICT) has been charged as one of the main contributors for organisational flattening [1]. The information revolution has given birth to new economies structured around processes and flows of data, information, and requirements. Thus, IT plays a significant role in the enabling or inhibiting of business process behaviour. Afsarmanesh and Camarinha-Matos [2] draw our attention to the emergence of Virtual Breeding Environments (VBEs) to support collaborative networks on a global scale discussing how business

networking opportunity exists on two levels. One level may be identified within the market while the other level may already exist within the VBEs. The growth of digital networks, electronic communication, e-commerce, Web technologies and other distributed systems has permitted the development of business-to-consumer and business-to-business relationships. This has opened up greater opportunities to cooperate on larger scales (i.e. the virtual organisation). Armed with opportunities such as greater connectivity and a free market, management continue to search for methods to maintain equilibria within their organisations. They search for the right balance or requisite variety between 'use, usage, and usability' of their resources and processes through diagnosis-oriented approaches [3]. On a virtual scale, this is proving to be extremely difficult.

A plethora of distributed systems across e-markets and e-services have emerged to offer to amalgamate network services on demand [4]. This raises important questions in regard to modern trajectories of workplace structures. For example, as organisational networks grow, how can we manage and monitor dispersed business processes? As organisational networks grow, so too do service networks to support business processes. These became known as distributed systems which interacted with one another and whose primary objective is to achieve a common goal, which is what happens within for example, Web services. Web services practitioners seek increased standardisation, openness, and integration. A Web service may be simply viewed as an application which is accessible to many other applications via the Internet and standard-based interfaces. The ultimate goal of Web services is inter-company through its business processes [5], and therefore should be easily 'defined, described, and discovered' (W3C). Many e-markets, may be considered as a system which facilitates the electronic exchange of information and business intelligence to conduct one or several business transactions, are sustained through the relationships made available through several Web services

II. BACKGROUND

Today, organisations are now forced to adapt their processes at a much faster pace than they have experienced before [6], and must also be proactive, reactive, and more decisive at a much faster rate than

ever before. Organisations are surrounded by the need to frequently change to meet consumer needs and failing to do so will put the organisation at risk. Thus, strategic and goal orientated management of technology is essential to reduce the probability of failure [7]. Assessing the value of the IT-enabled business processes is even of more critical importance and reveals how an organisation is positioned within a much larger network. Managing business processes (for example, measuring, monitoring, and changing) are essential activities across distributed business applications. According to [7], many companies fail because they cannot manage these fundamental factors successfully. There are many macro changes witness throughout today's business world due to economic, technological, or consumer-driven requirements. Papazoglou [8] draws our attention to a more micro level and explains that even a business transaction is a "consistent change in the state of the business that is driven by a well-defined business function" (p. 52). Web services are transcending from a 'promising technology' to becoming widely used within industry [8]. Papazoglou et al., [9] suggests that service technologies are being shaped by, and will shape modern services required in society (e.g. health, education, and business). Thus, Web services are becoming an integral and increasingly more integrated system into a much larger and complex distributed systems.

Gathering data on the health of organisational performance across a large network is very important. Organisations must attempt to shape data, information and knowledge if they want to strengthen their competitive position. Managing data (e.g. structural, functional, or behaviour) across Web services can be problematic. One of the major problems, as outlined by [10], is that managers are faced with a serious issue of how to manage "a completely invisible asset". Another cause highlighted by [11], is that managers have largely ignored the 'dynamic characteristics of networks and the ways that dynamic qualities of networks affect organisations' flexibility and change' (pp. 133). This has unavoidably led to organisations failing to capture the 'health' of their virtual presence, structure and workflows within business processes.

In recent years there has been significant interest in our ability to manage and engineer services. Organisations are beginning to move away from the traditional corporate hub of business practice towards a more diffused and distributed web of relationships and agile alliances. Across academia and industry we are beginning to recognise the significance of service innovation and service systems within the global 'service' economy. Understanding the mechanisms and theory of managing service business models and processes adopted across many organisations required the recent introduction of the Service Science Management and Engineering (SSME) discipline. However, it is clear that managers continue to face serious issues in managing 'a completely invisible asset' which inhibits their ability to monitor and exploit the value of agile service networks (ASN). Understanding the functionality of these networks and

the challenge of managing and co-coordinating their relationships is becoming more complex. As SSME theory is at a very early stage, it cannot prescribe methods to manage complex service processes and their relationships. The literature indicates that there is little emphasis on service discovery and design. Understanding the value of these relationships, especially from a technological perspective can prove to be extremely problematic. In addition, tools to create, track, and manage outsourcing business process opportunities are incompatible, slow, and difficult to use. It is also reported throughout literature that critical business data is not properly collected, shared, standardised, or analysed to provide business intelligence. To understand ASN affordances, we present a conceptual scenario of modelling web-enabled business process using social network analysis (SNA). This research borrows from SNA theory to present a new approach to monitoring and managing service networks. This method simulates future attention towards network behaviour and capability.

III. AGILE ORGANISATION

Nowadays, organisations are forced to adapt their processes at a much faster pace than before [6]. Managers must also be proactive, reactive and more decisive at a faster rate than ever before. Agility is a concept which attempts to address some of these issues. Defined by [12], agility is "an innovative response to an unpredictable change". It is concerned with taking greater control of unpredictable changes. Conboy and Fitzgerald [13] offer a definition of agility as: "the continual readiness of an entity to rapidly or inherently, proactively or reactively, embrace change, through high quality, simplistic, economical components and relationships with its environment". However, to do so, it is critical that we can access the value of agile resources. Assessing the value of the IT-enabled business processes reveals how an organisation is positioned within service network and its agile capability.

IV. KEY PERFORMANCE INDICATORS

KPIs are quantifiable measures of an organisational progress to meet specific goals. KPIs also assist managers in decision making to determine the right course of action. Within this digital age, information on organisational performance is of critical importance today. Information overload is one expression for the vast amounts of inundated data presenting managers with huge task to extract key information. Accessing the right information at the right time instead of taking time to retrieve information on organisational health is an important part of the overall process. Sifting through departmental and cross-organisational conflicting objectives clutters manager's ability to extract key performance information [14]. Freeing up resources to develop value-added information is critical to managerial activities (e.g. rapid decision making and execution). Clearly, grouping certain business process activities towards a focused objective within a VBE facilitates an improved market penetration strategy. Afsarmanesh and Camarinha-Matos [2] illustrate the

VBE life cycle stages which comprise of a number of phases in which a virtual network is formed (figure 1).

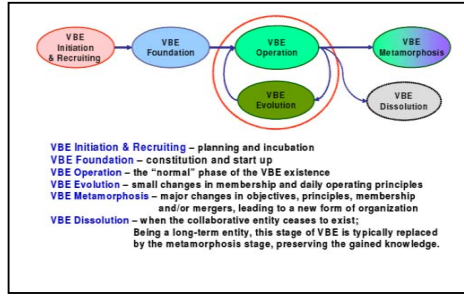


Figure 1. VBE Life Cycle Stages (extracted from [2]; p. 41)

To design management information systems which facilitate the VBE phases depicted above, we must first identify the KPIs within each phase. Many organisations become members of business transactions which facilitate the execution of numerous business transactions across a vast network of business applications. Monitoring Web services networks can allow us to develop software methods to enable us to understand and learn how to perfect the collaboration of shared business processes through networks of Web services [8]. To address these issues we must uniquely define the business KPIs. KPIs allow us to measure the success of goal achievement and to generate insight to discover whether a Web service, or an integral link in the network, has achieved its objective and reveal how performance may be enhanced. Characteristically, KPIs should be simple for decision making, relevant to unique business models, present timely results, useful, and instant for actionable insights. They may also be adjustable and independent to each organisation or department which contributes towards the overall objectives of the service network. Determining cross-organisational behaviour involves qualitative behaviour analysis (across many dimensions such as structural, functional, and behavioural; for example, see [15], p. 11). According to [16], KPIs are quite limited in what they can present to manager or analysts for strategic direction, i.e. they present what happened. This has led to the slowly emerging concept of Key Insights Analysis (KIA). The concept of KIAs will be further explored in alignment with the research progression, i.e. how and why something on a network happened.

V. KEY PERFORMANCE INDICATORS OF AGILE SERVICE NETWORKS

Capturing KPIs of ASNs within a framework allows managers to visualise the relations of various actors within the virtual organisational network. S-Cube [17] explain that ASNs “comprises large numbers of long-running, highly dynamic complex end-to-end service interactions reflecting asynchronous message flows that typically transcend several organisations and span geographical locations”. The sequence of message flows is a critical element of the overall transaction procedures. Developing a framework to monitor KPIs presents us with a reusable tool to assist managers to measure behaviour within ASNs. Baresi et al., [18],

state that the ‘reusability of specifications and designs through pattern and framework based approaches’ is becoming a critical issue which needs to be addressed in workflow software development (p.48). Increasing levels of attention is being placed upon customer satisfaction, of which timely and customised services are the key concepts. Pedrinaci et al., [6] explains that there is an urgent need for deploying solutions able to adapt business processes in a more agile and less timely manner, guided by the existing specific objectives and organisational strategies. Afsarmanesh and Camarinha-Matos [2] explore the concept of VBE and report a number of factors in their design of a VBE framework, including competency management, value system, system of metrics, and trust management within the network. To understand the necessity to place value on networked organisation and technological infrastructures, we must begin to explore business processes through a new lens.

VI. BUSINESS PROCESS MANAGEMENT

Business processes management (BPM) has emerged as one of the major new developments within organisations to support our understanding of the evolution and interaction of process-oriented business applications and information systems. According to [8], a business process is “a set of logically related tasks performed to achieve a well defined business outcome” (p. 49). BPM has encapsulated many definitions over time, which identified the need to enhance processes and allow organisations to operate more efficiently. Elzinga [19] state that BPM consists of, “...systematic, structured approaches to analyse, improve, control, and manage processes with the aim of improving the quality of products and services.” Zairi [20] describes BPM as, “...a structured approach to analyse and continually improve fundamental activities such as manufacturing, marketing, communications and other major elements of a company’s operations”.

There is a presumption within BPM that processes are continuously improved by analysing and implementing necessary changes within them. Hepp et al., [21], definition is more fitting when discussing business process on a virtual scale. They report that BPM aims at, “...providing tools and techniques that support the modelling, management, and monitoring of operations on a business process level, while automatically mapping this high-level perspective to the actual implementation being executed on the multiplicity of systems.” This definition introduces the view of a much wider network, i.e. across a number of departments which is responsible for business process management success. Lindsay et al., [22] introduce the ability to change and improve organisational performance and see BPM as an attempt to “...better understand a business’s key mechanisms in order to improve, and in some cases radically change, the business performance by identifying opportunities for new business opportunities, for outsourcing, for improving business efficiency and for areas within the business where technology can be used to support business processes”. Here we see BPM extending outside of the organisation, and outsourcing processes

to improve efficiency. As more and more organisations adopt this approach, many organisations form a virtual collaborative network of alliances to which many business processes are outsourced. Business processes are usually dynamic, and of lengthy periods of time, which involves harmonisation across numerous manual and automated tasks, and require access to several different applications and the interpretability of several application systems governed by complex business rules [23]. The ultimate goal of Web services is to automate the execution of business processes across inter-company and standardised business applications. Business process modelling describes the structure and behaviour of organisational activity (workflow, roles, and rules). Therefore, it is inevitable that companies must find new strategies and performance insights to achieve competitive advantages through BPM. One of the major emphases realised today in achieving a competitive advantage is in business intelligence (BI), through communicative and collaborative networks and knowledge management (KM) across the wider organisational spectrum. Wetzstein et al., [24] explore the use of Business Activity Monitoring to map service choreography and monitoring service processes. However, developments in BI, BPM, and Business Activity Monitoring (BAM) to enhance less tangible tasks across ASN have not been very successful in the past. To assist business analysts and managers to extract knowledge, there are a number of BI tools available (e.g. IBM Cognos 8 is a single service-oriented architecture). However, these are limited in their functionality (e.g. ability to extract unstructured data, on a single organisation). Pedrinaci et al., [6] and [25] explains that there is an urgent need for deploying solutions capable of deriving more sophisticated insights of business processes. The ability to quickly extract accurate data from ASN presents business with more opportunity to gain competitive advantage and identify automated agile value of business processes within their cross-organisational network. Papazoglou [8], describes an automated business process as a “precisely choreographed sequence of activities systematically directed towards performing certain business tasks to completion” (p. 49). Papazoglou [8] adds that a business process choreography consists of “the aggregation of Web services by encoding business rules or patterns governing service interactions and has the ability to reuse the created aggregations” (p. 51). Understanding the business logic of Web services is an initial attempt to extract KPIs and learn how we can implement a framework to govern service behaviour across ASN.

VII. WEB SERVICES

Hung et al., [26] define a Web service as “an autonomous unit of application logic that provides either some business functionality or information to other applications through an Internet connection” (p. 1-2). The Service-Oriented Computing (SOC) paradigm extends this concept to ‘the set of concepts, principles, and methods that represent computing in Service-Oriented Architecture (SOA) in which software applications are constructed based on

independent component services with standard interfaces’ [27]. SOA is also looked upon as a method which allows businesses to ‘contract-out their noncritical functions’ [28]. Papazoglou et al., [29] calls for a more holistic approach to Service SOC research, “to unleash the full potential”. Papazoglou et al., [29] report that, “the visionary promise of Service-Oriented Computing is a world of cooperating services where application components are assembled with little effort into a network of services that can be loosely coupled to create flexible dynamic business processes and agile applications that may span organisations and computing platforms” (p. 225). The Internet, World Wide Web, and more recently, Web services, afford us the transparent access to services in distributed and heterogeneous environments [29]. Gottschalk et al., [25] explains how the Internet evolved from the business-to-customer (B2C; “eyeball web”) to a business-to-business (B2B; “transactional web”). A Web service may be defined as a set of technologies which dynamically support interoperable interaction to exchange data between applications over the Internet for business functionality. These activities must cross ‘checkpoints’ which represent business conditions and rules [8]. Many businesses are now taking advantage of the Internet infrastructure, allowing Web services to become a popular method to extend business application-to-application interaction. Each application which is interacted with becomes an important link of a Web service. The Web service framework is divided into three main categories; communication protocol, service description, and service discovery. Web Services Management (WSM) is becoming a mandatory task for organisations that enter the online economy. This surmounts to an organisations capacity to distinguish and communicate these new business models to improve focus, establish an agile framework, and position an organisation to thrive despite industry discontinuities [30]. As organisations face increased levels of competition, many opt for the new ‘online economy’, to offer their agile business competencies through Web services [31]. Goldman et al., [32] states that agility is dynamic, context-specific, aggressively change-embracing, growth-orientated, success in competitive arenas, winning profit and market shares. Organisational agility is of greater importance today and is viewed as a good strategy to ‘survive and thrive in turbulent times’ and offers more profitability [14]. Glenn [14] reports that most organisations are inflexible, face internal barriers, and lack supporting technologies to encourage organisational agility. Due to the use of standard protocols, Web services enjoy a sophisticated level of interoperability among many applications. Interoperability refers to a property of diverse systems across a network while organisations enabling them to work together to achieve a measurable outcome. The design of Web services must clearly reflect a business model which encapsulates the sequence of agile business processes to provide value to customers demand. According to [5], a “web services work on the assumption that the functionality made available by companies will be exposed as a service...a service is a procedure, method, or object

with a stable, published interface that can be invoked by clients.....Thus requesting a service involves a program calling another program” (pp. 131). Agile methods have received increasing levels of attention in recent years. According to [33] and [29], agile organisations require new forms of organisational structure, non-hierarchical management structures, stimulate and support individuals, cooperation and team working, and the need for advanced computer based technologies. One of the major obstacles within this domain is to understand the value of IT-enabled behaviour across an ASN.

VIII. SOCIAL NETWORK ANALYSIS

We selected social network analysis (SNA) from a number of techniques including benchmarking, game theory, knowledge mapping, and mining techniques (for example, see [15]; p. 8). SNA is an approach and set of techniques which studies the exchange of resources (for example, information) among actors. SNA focuses on patterns of relations among nodes such as people, groups, organisations, or information systems ([34]; [35]; [36]; [37]). SNA demonstrates the ties and relationships between each node to provide a visual and mathematical representation of interaction and exchanges which influence behaviour. Hassan [38] adopts a more information systems view and defines an actor in SNA as, “...any object, individual or group of objects or individuals that are involved in the network of relations”. Managers realise that the key to continued success is within their understanding of how workflows and business processes can be optimised (e.g. [39]. Balkundi and Kilduff [40], report that SNA may allow organisations in financial trouble to gain vital insights and discover survival prospects. Thus, additional focus should be placed on tailoring business processes to guide and support the discovery of KPIs (and process patterns). Kawalek and Greenwood [41], describes an abstract model of an organisation, and how we can develop our understanding through the addition of three models:

1. A model of the system: a high level, structural view of organisational interactions (who and/or what interacts).
2. A model of goals: having identified patterns of interaction in the model, how can we describe the interactions (why do they take place).
3. A model of methods: having identified what interacts and why, a model is developed to determine why and how goals are achieved.

As outlined above, few studies explores IT interactions and its contribution towards business processes. To add a fourth step to [41] abstract model, from an agile perspective, it would be extremely useful to implement a ‘model of action’, i.e. a model which would allow us to explore strategic possibilities to simulate a ‘what-if’ approach to understanding the influence of each relationship across business processes. This appears to be simply taken for granted throughout literature. However, [38] demonstrates that by studying IT-enabled processes, we can identify the contribution of IT to business process success, or

improved performance. SNA is an interdisciplinary approach, now being adopted to solve many traditional business and managerial problems. Cross et al., [42] state that researchers using SNA in management studies ‘have clearly demonstrated the extent to which informal networks pervade and effect life and work within organisations’ (p. 5). In addition, [43] describes SNA as a method for detecting, describing, and analysing the relationships among a group of people or organisations, thus the focus is not on individuals as discrete units of analysis, but rather on the relationship that exist within the group. Another benefit of SNA is its ability to provide a methodology to gain deeper insight of how structural regularities influence behaviour [44]. SNA assumes that actors are interconnected, with real consequences for behaviour and performance. Structures may be altered to optimise the networks outcomes. Therefore, SNA is a very fitting methodology to deploy within this research to uncover more ‘truths’ as to the activities and their patterns within agile service networks.

IX. CONCLUSION AND FUTURE RESEARCH

This paper provides a survey of the literature and introduces the main concerns within business process management and service science. It has successfully identified some of the key problems, and places greater emphasis on the need to introduce methods to visualise agile service networks across virtual organisations. To do so, the researcher has selected a research methodology of SNA which will guide the research in achieving its objectives. The expected outcome of this research is to present a concrete and practical framework which will empower managers to perform ad-hoc analysis of Web-enabled business processes across ASN. The practicality of this framework should also assist managers who are participating in a virtual agile business environment, to assess, manage, or restructure these networks. This will also test the validity of the framework. Therefore we expect three explicit benefits from this research. Firstly we address the gap in literature within our ability to monitor and map business processes across agile service network. Secondly, this research will present a practical framework which can be utilised (and reused) in industry to address some concerns in agile networks while extending the SSME body of knowledge, e.g. service management. Thirdly, it will demonstrate how the SNA methodology can be a useful tool to managers as a performance mechanism.

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